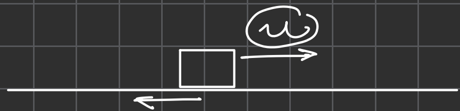
FRICTION.

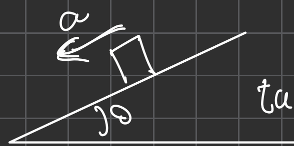
Recap.



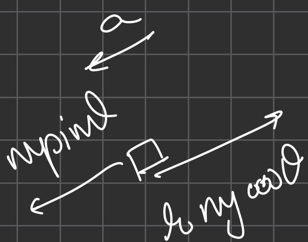
1) $f_k = \mu N = \mu mg$ ← kinetic friction

2) $f_{static} \le f_{Limiting} \; (\mu N)$

3)



$\tan\theta > \mu$

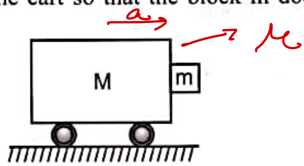


$a = g\sin\theta - \mu g\cos\theta$

Doubts

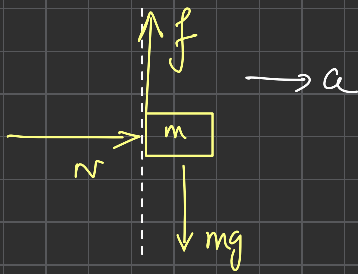
Ex I Q : 20

20. A cart of mass M has a block of mass m placed against it as shown in the figure. Co-efficient of friction between the block and cart is μ. What is the minimum acceleration of the cart so that the block m does not fall?



(a) μg (b) μ/g

(c) g/μ (d) none



$$\Sigma F_x = ma$$

$$\Rightarrow N = ma \quad \text{——} \quad (1)$$

$$\Sigma F_y = 0$$

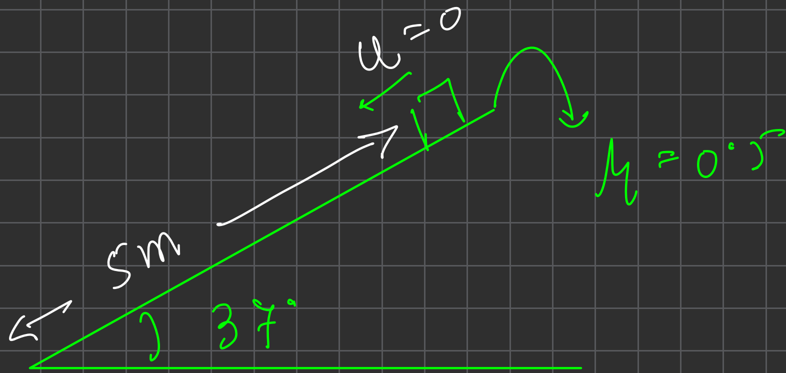
$$\Rightarrow f = mg \quad \text{——} \quad (2)$$

For limiting case

$$\mu N = mg$$

$$\Rightarrow \mu(\cancel{m}a) = \cancel{m}g \Rightarrow a = g/\mu$$

Q.1)



Time for block to slide down incline.

A) 1 s   B) $\sqrt{2}$ s

C) $\sqrt{3}$ s   D) $\sqrt{5}$ s

Ans → $\tan\theta = 3/4 = 0.75 > \mu(0.5)$

$\Rightarrow$ slide

→ $a = g(\sin\theta - \mu\cos\theta)$

$= 10\left(3/5 - \frac{1}{\cancel{2}} \times \frac{\cancel{4}^{2}}{5}\right) = 10 \times \frac{1}{5} = 2$

→ $S = 5$

→ $u = 0$

$\underbrace{s = \frac{1}{2}at^2}$ $\Rightarrow$ $t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2\times 5}{2}}$

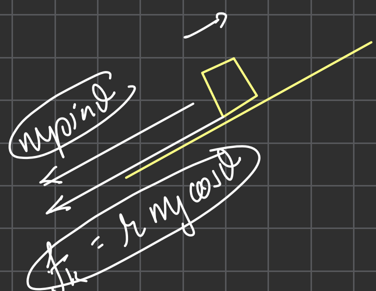
$\Rightarrow$ $\boxed{t = \sqrt{5}}$

Q.2)

Find u such that block moves by 5m before turning.

A) 10 m/s B) 5 m/s

C) 4 m/s D) 6 m/s.



$$a = -\frac{(mg\sin\theta + \mu mg\cos\theta)}{m}$$

$$= -g(\sin\theta + \mu\cos\theta)$$

$$= -10\left(\frac{3}{5} + \frac{1}{2}\times\frac{4}{5}\right)$$

$$= -10 \text{ m/s}^2$$

$s = 5$

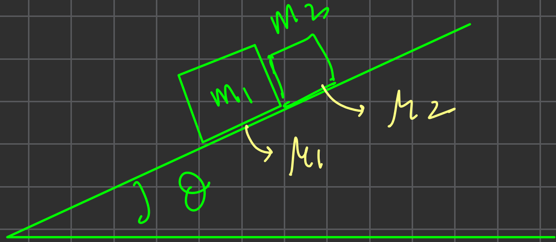
$v = 0$

$u = ??$

$v^2 = u^2 + 2as \Rightarrow 0 = u^2 - 2\times 10\times 5$

$\Rightarrow u^2 = 100 \Rightarrow \boxed{u = 10}$

Q3.



$$\boxed{\begin{aligned} \tan\theta &> \mu_1 \\ \tan\theta &> \mu_2 \end{aligned}}$$

$\downarrow$

Both blocks slide

→ If $\underline{\mu_2 > \mu_1}$ nature of motion ??

Ans: μ_1 less ⇒ smoother ⇒ higher a of m_1

or $a = g\sin\theta - \underline{\mu g\cos\theta}$ ⇒ $\mu\downarrow\ a\uparrow$

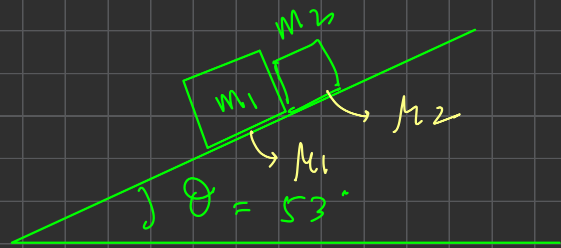
as $a_1 > a_2$ ⇒ blocks move separately

If $\underline{\mu_2 \le \mu_1}$ nature of motion ??

Ans: ⇒ a of $m_2 > m_1$ (if independent)

⇒ blocks move together (as not possible for m_2 to move ahead)

Q.



$m_1 = 2\,kg$

$m_2 = 3\,kg$

$\mu_1 = 0.2$

$\mu_2 = 0.1$

Find 1) a of blocks

2) 'N' between blocks.

1) A) $4.2\ m/s^2$ B) $7.2\ m/s^2$

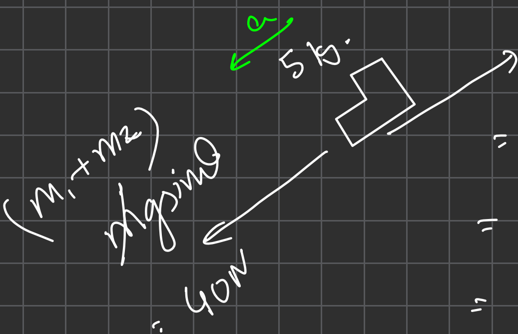
C) $6\ m/s^2$ D) $5\ m/s^2$

2) A) $0.7\ N$ B) $1.4\ N$

C) $2.8\ N$ D) $4.2\ N$

1) $\tan\theta = 4/3 = 1.33 > \mu \Rightarrow$ slide

$\mu_1 > \mu_2 \Rightarrow$ move together



$f_1 + f_2 = \underline{4.2N}$

$= \mu_1 m_1 g\cos\theta + \mu_2 m_2 g\cos\theta$

$= (0.4 + 0.3)\underbrace{\boxed{g\cos\theta}}_{6}$

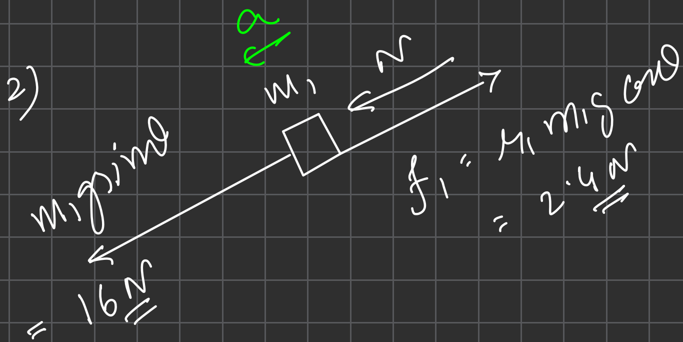
$= 0.7 \times 6$

$= 4.2\ N$

$$\Sigma F = ma$$

$$\Rightarrow a = \frac{40 - 4.2}{5} = \frac{35.8}{5} = 7.16 \, m/s^2$$

2)



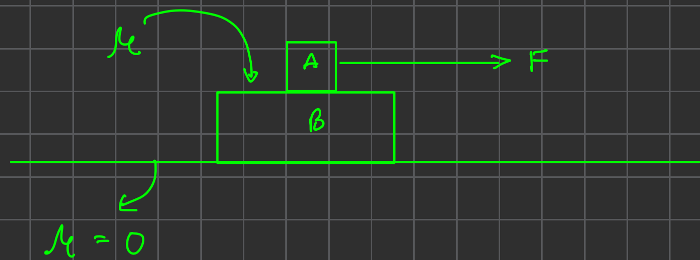
$$\Sigma F = ma$$

$$\Rightarrow 16 + N - 2.4 = 2 \times 7.16$$

$$\Rightarrow N = 14.32 + 2.4 - 16$$

$$= 0.72$$

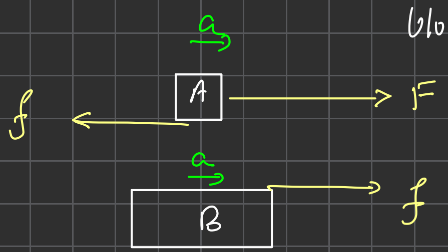
Block on block problems.



Possibilities.

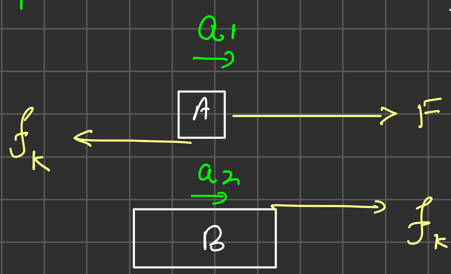
1) Both move together. No slipping between A & B.
2) A slips on B

Case I Both move together. Nature of friction between blocks = static f



[To find direction of friction
1) Assume no friction & find direction of relative motion
2) friction acts opposite to that)

Case II A slips on B ⟶ Kinetic friction, different accelerations



Steps to solve two block problems

1) **Step one**, assume both the blocks move together and find the common acceleration

2) **Step 2** draw individual FBD's and find the static friction between the two blocks

3) **Step 3** Compare static f with limiting f.

If $\boxed{\text{static } f \le f_{\text{limiting}}}$

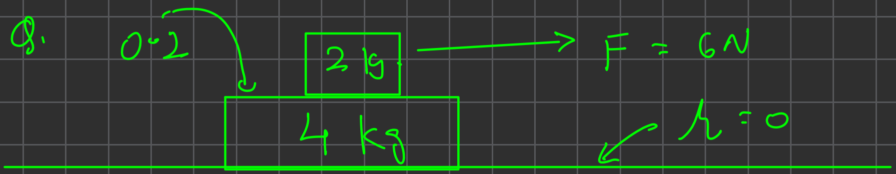
$\Rightarrow$ both blocks move together.

4) **Step 4** If $\boxed{\text{static } f > f_{\text{limiting}}}$

$\Rightarrow$ don't move together $\Rightarrow$ slipping

$\Rightarrow$ apply kinetic friction, assume separate accelerations and solve.

Q.



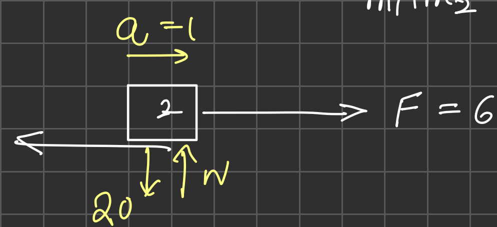
Find 'a' of blocks & friction between them.

Ans: Step 1

Assume साथ / together.

$$a = \frac{F}{m_1+m_2} = \frac{6}{6} = 1\ m/s^2$$

Step 2



$$\Sigma F = ma$$

$$\Rightarrow 6 - f = 2\times 1 \Rightarrow f = 4N$$

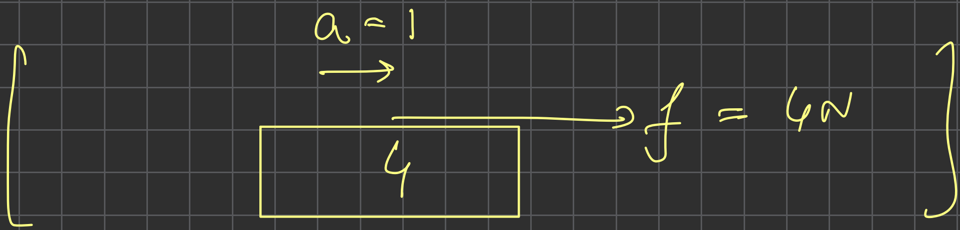
Step 3

$$f_{limiting} = \mu N$$
$$= 0.2 \times 20 = 4N$$

$\therefore$ $f_{static} \le f_{limiting}$

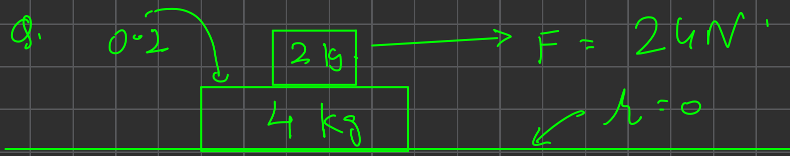
$\Rightarrow$ together.

$$\therefore \boxed{a = 1, \quad f = 4}$$



friction on lower block causes it to accelerate.

Q.



Find 'a' of blocks & friction between them.

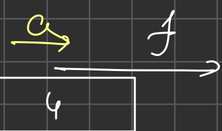
Ans: Step 1

F = 24N

$$a = \frac{24}{6} = 4\,m/s^2$$

Step 2 Calculate static f.

FBD of any 1 block.



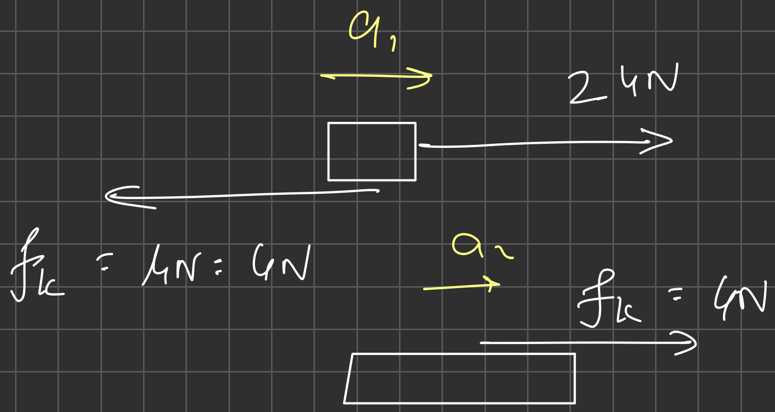
$$\Sigma F = ma \Rightarrow f = 4 \times 4 = 16\,N$$

Step 3 Compare with limiting f

$$f_L = \mu N = 0.2 \times 20 = 4N$$

As static > f_L => slipping.

Step 4 Solve using kinetic friction



$$a_1 = \frac{\Sigma F}{m_1} = \frac{24-4}{2} = \underline{\underline{10\, m/s^2}}$$

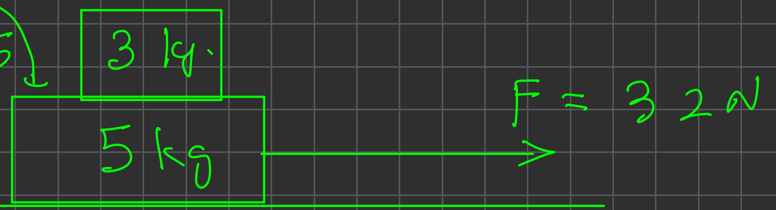
$$a_2 = \frac{\Sigma F}{m_2} = \frac{4}{4} = \underline{\underline{1\, m/s^2}}$$

$$\therefore \boxed{\begin{array}{l} a_1 = 10 \\ a_2 = 1 \end{array} \quad / \quad f = 4N}$$ Answer

---

Q. $\mu_s = 0.5$

$\mu_k = 0.4$



$\mu = 0.$

'a' of blocks, friction ??

'a' ~~A)~~ 4, 4 B) 4, 6.4

C) 4, 3.4 D) None

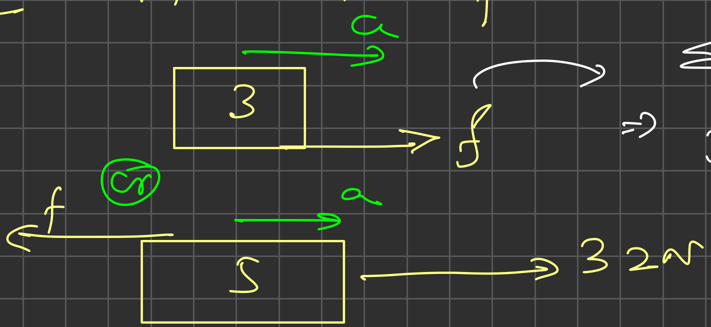
f A) 15 N ~~C)~~ 12 N

C) 25 N D) 20 N

Step 1 Assume together

$$a = \frac{F}{m_1 + m_2} = \frac{32}{8} = \underline{4}$$

Step 2 Find static f

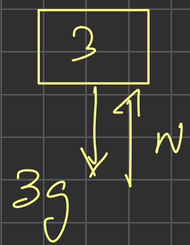


$\Sigma F = ma$

$\Rightarrow f = 3 \times 4$

$= \underline{\underline{12N}}$

step 3 Compare with $f_{limiting}$.



$\therefore N = 3g = \underline{\underline{30}}$

$\therefore f_L = \mu_s N = 0.5 \times 30$

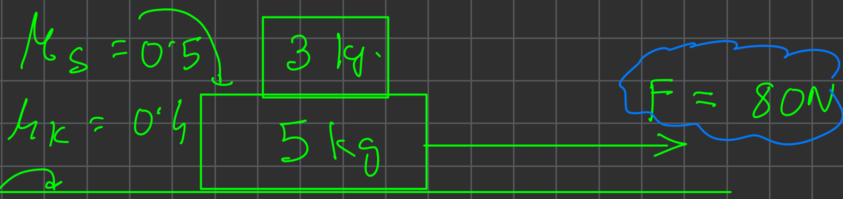
$= \underline{\underline{15N}}$

As static f (12N) $<$ f_L (15N)

$\Rightarrow$ move together

$$\boxed{\begin{aligned} \therefore a_1 &= a_2 = 4 \\ f &= 12N \end{aligned}}$$

Q. Repeat above Q. with $F = 80N$.



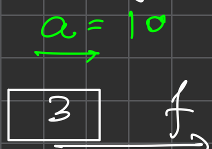
$\mu = 0$.

'a' of blocks, friction ??

☆(If slipping, use $\mu_{kinetic} = 0.4$)

Ans: 1) Assume together. $a = \frac{80}{8} = 10\ m/s^2$

2)



$\Rightarrow \Sigma F = ma$

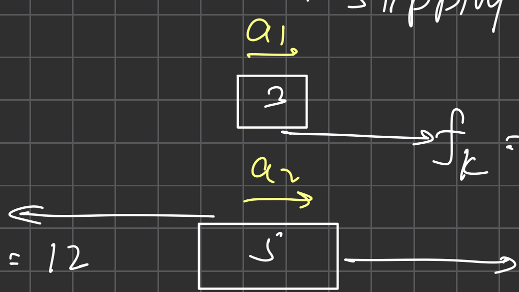
$\Rightarrow f = 3 \times 10 = 30$.

3) compare with $f_L = \mu_s N = 0.5 \times 30 = 15N$

As static $f > f_L$

$\Rightarrow$ slipping $\Rightarrow$ kinetic f acts

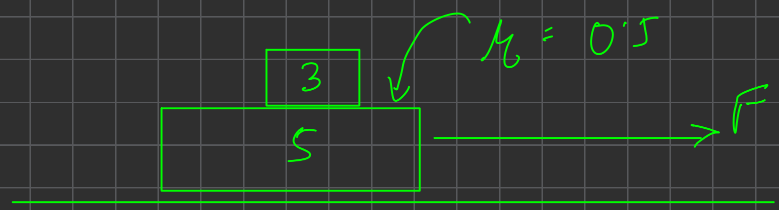
4)



$f_k = \mu_k N = 0.4 \times 30 = 12$

$$\begin{cases} a_1 = \dfrac{12}{3} = \underline{\underline{4\,m/s^2}} \\[2ex] a_2 = \dfrac{80 - 12}{5} = \dfrac{68}{5} = \underline{\underline{13.6\,m/s^2}} \\[2ex] f_k = 12\,N \end{cases}$$

Q.



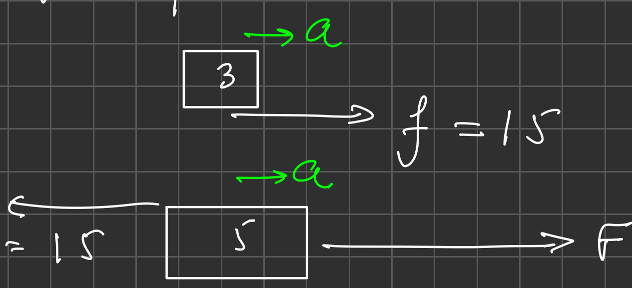
what can be maximum force F so that both move together.

☆ Max. F (Boundary), $f = f_{limiting}$.

A) 15N B) 25N

C) 40N ✓ D) 50N

Ans. $f = f_{limiting} = \mu N = 0.5 \times 30 = 15N$



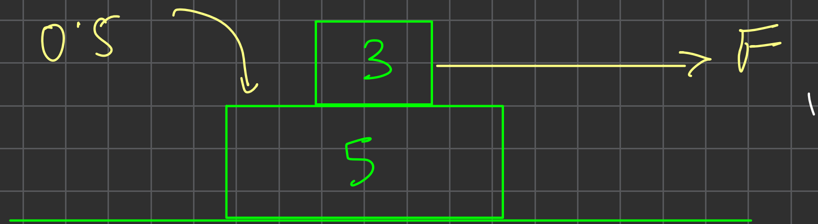
3 kg. $15 = 3a \Rightarrow a = 5$ — (1)

5 kg. $F - 15 = 5a$
$= 25$

$\Rightarrow F = 40N$

Answer.

Q.

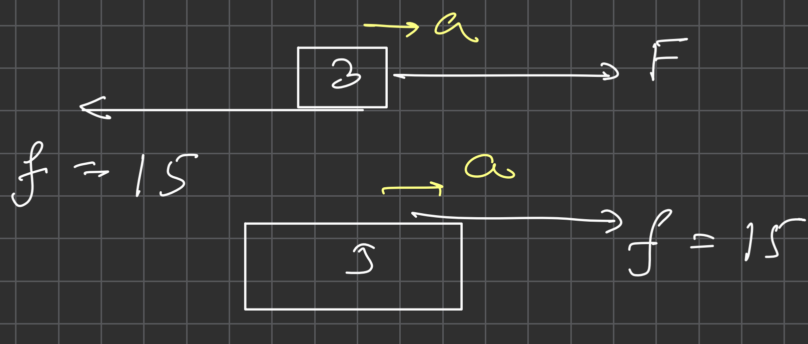


Find max. F so that blocks move together.

~~A)~~ 24 B) 32

C) 40 D) 48

Ans: $f = f_{limiting} = \mu N = 0.5 \times 30 = 15$

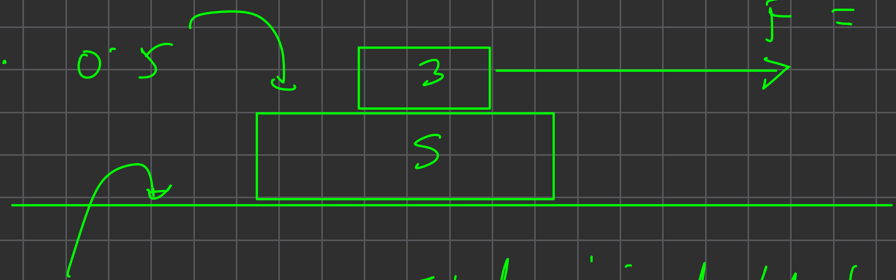


3 kg: $F - 15 = 3a$ —— (1)

5 kg: $15 = 5a \Rightarrow \underline{a = 3}$.

$\therefore F - 15 = 3 \times 3 \Rightarrow \boxed{F = 24N}$

Q. 0.5 $\quad F = 16N$

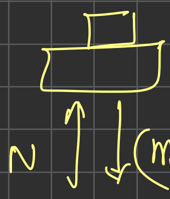


$\mu = 0.1$

Find 'a' of blocks & f.

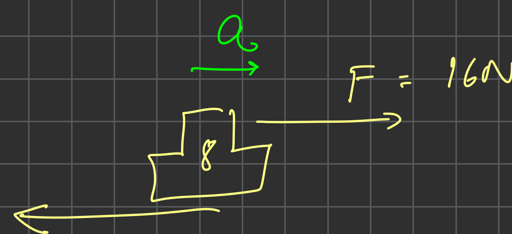
Ans: ground → $f_{limiting} = \mu N$ ground

$= 0.1 \times 80 = 8N$



N, $(m_1+m_2)g = 80$

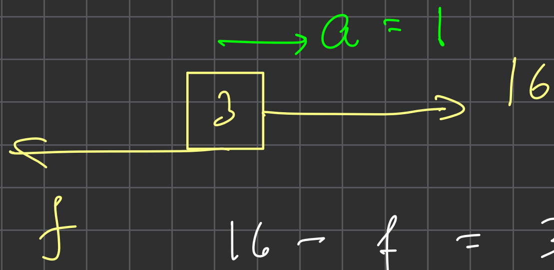
1) Assume together.



$f_K = 8N$

$$a = \frac{16 - 8}{8} = 1\ m/s^2$$

2) Find f between blocks.



$16 - f = 3 \times 1$

$\Rightarrow \underline{\underline{f = 13N}}$

3) $f_{limiting} = \mu N = 0.5 \times 30$

$= \underline{\underline{15N}}$

As static $f < f_{limiting}$

$\Rightarrow$ together.

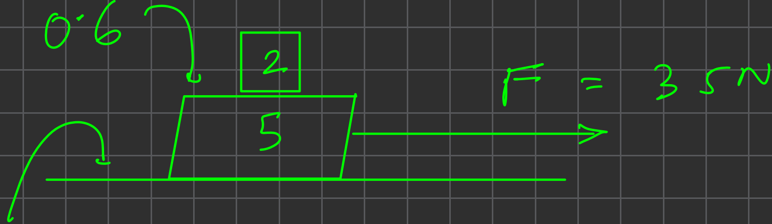
$$\boxed{\begin{aligned} a_1 &= a_2 = 1\,m/s^2 \\ f &= 13N \end{aligned}}$$

Ans.

---

* In above problem, blocks move only if $F > 8N$ (ground friction).

Q.



$\mu = 0.2$

Find 'a' of blocks & f between them.

a: A) 0, 0 B) 5, 5

~~C)~~ 3, 3 D) 6, 7

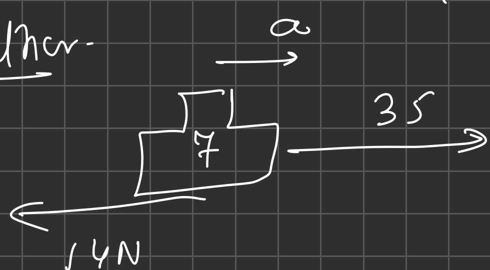
f: A) 10 ~~B)~~ 6

C) 12 D) 4

Ans: friction by ground = $\mu N = 0.2 \times 70$

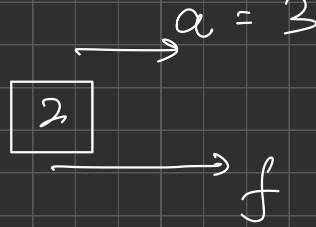
$= 14N$

i) Assume together.



$$a = \frac{35 - 14}{7} = 3\ m/s^2$$

2) Find f static



$\therefore f = 2 \times a \Rightarrow f = 6N$.

3) $f_{limiting} = \mu N = 0.6 \times 20 = 12N$.

$\therefore \boxed{\text{static } f < f_{limiting}}$

$\therefore$ move together.

$$\boxed{\begin{array}{c} a_1 = a_2 = 3 \\ f = 6N \end{array}}$$

Answer.

CALCULUS.

Differentiation

- Formulae (Maths.)
- Interpretation
  - Normal
  - graphically
- Application.

Rules / Formulae

1) Function

→ y = function of x (or) $y = f(x)$

⇓

value of y depends on x.

Eg. $y = 2x + 1$

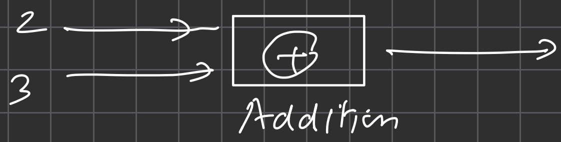
$$\begin{cases} x = 0 \Rightarrow y = 2\times 0 + 1 = 1 \\ x = 1 \Rightarrow y = 2\times 1 + 1 = 3 \\ x = 4 \Rightarrow y = 2\times 4 + 1 = 9. \end{cases}$$

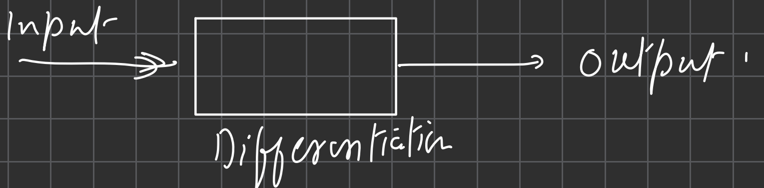
$\vdots$

Eg. $y = \sin x.$

$$\begin{cases} x = 0 \Rightarrow y = 0 \\ x = \pi/6 \Rightarrow y = \sin \pi/6 = 1/2 \\ x = \pi/3 \Rightarrow y = \sin \pi/3 = \sqrt{3}/2 \ldots \end{cases}$$

2) Differentiation → Mathematical operator





Notation: $\boxed{\dfrac{dy}{dx}}$ or y'

→ Differentiation of y with respect to x

($y = f(x)$ or some function of x)

Eg. $\dfrac{d(\sin x)}{dx}$, $\dfrac{d(2x+1)}{dx}$ etc.....

we are differentiating '$\sin x$' w.r.t x.

* (Differentiation = derivative)

Formulae/ Rules

1) Power rule $y = x^n$ (n = any real no.)

$$\boxed{\frac{dy}{dx} = \frac{d(x^n)}{dx} \longrightarrow n\,x^{n-1}}$$

Q1. $\frac{d}{dx}(x^2) \xrightarrow[n=2]{} n\,x^{n-1}$

$= 2\,x^{2-1} = 2x$

$\therefore$ Diffⁿ of $x^2 = 2x$.

Q2. $\frac{d}{dx}(x^4) \xrightarrow[n=4]{} 4\,x^{4-1} = 4x^3$

Q3. $\frac{d}{dx}\,x^{3/2} \xrightarrow[n=3/2]{} \frac{3}{2}\,x^{3/2-1} = \frac{3}{2}\,x^{1/2}$

$= \frac{3}{2}\sqrt{x}$.

Q4. $\frac{d}{dx}\sqrt{x} \xrightarrow[n=\frac{1}{2}]{} \frac{1}{2}\,x^{1/2-1} = \frac{1}{2}\,x^{-1/2}$

$= \frac{1}{2\,x^{1/2}} = \frac{1}{2\sqrt{x}}$

Q5. $\frac{d}{dx}\left(\frac{1}{x}\right)$ (x^{-1}) $\xrightarrow[n=-1]{} (-1)\,x^{-1-1} = -\frac{1}{x^2}$

Q6. $\frac{d}{dx}(x) \xrightarrow[n=1]{} 1\,x^{1-1} = 1\,x^0$

$= 1$

2) If y = constant (c)

$$\boxed{\frac{d(c)}{dx} = 0}$$

Differentiation of constant is 0

0, 1, 2, 3, π, $\sqrt{2}$, e etc.

Eg. $\frac{d}{dx}(3^4) = 0$

3) Constant multiplication rule

$$\frac{d(cy)}{dx} = c\,\frac{dy}{dx}$$

Eg. $\frac{d}{dx}(4x) = 4\,\frac{d(x)}{dx} = 4 \times 1 = 4$

Eg. $\frac{d}{dx}(3x^5) = 3 \times \frac{d}{dx}(x^5)$

$= 3 \times 5x^{5-1} = \underline{\underline{15x^4}}$

4) Addition / subtraction rule

$$\frac{d}{dx}(y \pm z) = \frac{dy}{dx} \pm \frac{dz}{dx}$$

Eg. $\frac{d}{dx}(x^2 + 4x + 1) = \frac{d}{dx}(x^2) + \frac{d}{dx}(4x) + \frac{d}{dx}(1)$

$= 2x + 4 \times 1 + 0$

$= 2x + 4$

Eg. $\frac{d}{dx}(4x^3 - 6) = \frac{d}{dx}(4x^3) - \frac{d}{dx}(6)$

$= 4 \times 3x^2 - 0$

$= 12x^2$

5) Trigonometric

	$f(x)$	dy/dx
1)	$\sin x$	$\cos x$
2)	$\cos x$	$-\sin x$
3)	$\tan x$	$\sec^2 x$
4)	$\sec x$	$\sec x \tan x$
5)	$\cot x$	$-\operatorname{cosec}^2 x$
6)	$\operatorname{cosec} x$	$-\cot x \operatorname{cosec} x$

$$\left(\frac{d}{dx}(\sin x) = \cos x\right)$$

Find 1) $\frac{d}{dx}(5\cos x + 2\tan x)$

$$= 5(-\sin x) + 2(\sec^2 x)$$

$$= -5\sin x + 2\sec^2 x$$

2) $\frac{d}{dx}\left(\frac{\tan x}{4}\right) = \frac{1}{4}\frac{d}{dx}(\tan x)$

$$= \frac{1}{4} \times \sec^2 x$$

6) Exponential / logarithmic function

$$e^x \longrightarrow e^x \qquad (e = 2.718\ldots)$$

$$\log_e x \text{ or } \ln x \longrightarrow \frac{1}{x}$$

$$\left(\frac{d}{dx}(\ln x) = \frac{1}{x}\right) \quad \frac{d}{dx}(e^x) = e^x$$

7) Product rule

$(u, v = 2 \text{ functions of } x)$

$$\boxed{\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}}$$

Eg. $\frac{d}{dx}(x \sin x) = x\frac{d(\sin x)}{dx} + \sin x \frac{d(x)}{dx}$

$= x \cos x + \sin x$

Eg. $\frac{d}{dx}(x^2 \tan x) = x^2 \underbrace{\boxed{\frac{d}{dx}(\tan x)}}_{\sec^2 x} + \tan x \underbrace{\boxed{\frac{d(x^2)}{dx}}}_{2x}$

$= x^2 \sec^2 x + 2x \tan x.$

Eg. $\frac{d}{dx}(\tan x \cos x) = \tan x \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(\tan x)$

$= -\tan x \sin x + \cos x \sec^2 x$

Eg. $\frac{d}{dx}(4x e^x)$

$= 4 \frac{d}{dx}(x e^x)$

$= 4 \left[x \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(x) \right]$

$= 4 \left[x e^x + e^x 1 \right]$

$= 4e^x [x + 1]$

8) Division rule

$$\boxed{\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\left(\frac{du}{dx}\right) - u\left(dv/dx\right)}{v^2}}$$

Eg. $$\frac{d}{dx}\left(\frac{\sin x}{x}\right) = \frac{x\,\frac{d(\sin x)}{dx} - \sin x\,\frac{d(x)}{dx}}{x^2}$$

$$= \frac{x\cos x - \sin x \times 1}{x^2}$$

$$= \frac{x\cos x - \sin x}{x^2}$$

Eg. $$\frac{d}{dx}\left(\frac{x^2}{e^x}\right) = \frac{e^x\left(\frac{d}{dx}(x^2)\right) - x^2 \times \frac{d}{dx}(e^x)}{(e^x)^2}$$

$$= \frac{e^x(2x) - x^2(e^x)}{e^{2x}}.$$

Eg. $$\frac{d}{dx}\left(\frac{x^2+1}{\sin x}\right) = \frac{\sin x\,\frac{d}{dx}(x^2+1) - (x^2+1)\frac{d}{dx}(\sin x)}{\sin^2 x}$$

$$= \frac{\sin x(2x+0) - (x^2+1)\cos x}{in^2 x}$$

$$= \frac{2x\sin x - (x^2+1)\cos x}{\sin^2 x}$$

H.W. Friction →

Intext 3 → All (except 24, 30)

Ex 2 → 4, 8, 16, 17, 20 − 25, 31

DPP → 7.4 4 7.5

Differentiation

Intext 1 → 1, 2, 3, 7 − 10

Ex 1 ⇒ 1, 2, 7, 9, 11, 12, 13

* $\boxed{\text{Intext, Ex 1}}$ solved Q.

↓

Ex 2 / DPP.